DeadlineDinosaur: Integrating System-Aware Optimization with Hybrid Scheduling for the 60-Second 3DGS Challenge

Parag Sarvoday Sahu
IIT Gandhinagar
India
parag.sahu@iitgn.ac.in

Vishwesh Vhavle
IIT Jodhpur
India
vishwesh@iitj.ac.in

Kshitij Aphale
IIT Jodhpur
India
aphale.1@iitj.ac.in

Avinash Sharma
IIT Jodhpur
India
avinashsharma@iitj.ac.in

Abstract

The SIGGRAPH Asia 2025 3D Gaussian Splatting Challenge presents a distinct engineering constraint: maximizing reconstruction quality within a strict 60-second limit on a single GPU. This deadline necessitates a departure from standard optimization routines, prioritizing iteration density and resource allocation over asymptotic convergence. In this work, we present *DeadlineDinosaur*, a framework derived from a systematic integration of recent acceleration techniques. We utilize **LiteGS** as a computational backbone for its highly optimized warp-based rasterization. On this foundation, we layer the frequency-guided resolution scheduling of **DashGaussian** and the momentum-based budgeting of **Taming3DGS**. Our approach leverages low-resolution supervision in the early stages to maximize the total number of gradient updates possible within the time budget. This strategy secured the **2nd Rank** on the official challenge leaderboard, achieving an average PSNR of 28.04 with a runtime of 57 seconds, demonstrating the efficacy of hybrid scheduling in constrained scenarios.

CCS Concepts

• **Computing methodologies** → **Volumetric models**.

Keywords

3D Gaussian Splatting, System Integration, Real-time Optimization

ACM Reference Format:
Parag Sarvoday Sahu, Vishwesh Vhavle, Kshitij Aphale, and Avinash Sharma. 2025. DeadlineDinosaur: Integrating System-Aware Optimization with Hybrid Scheduling for the 60-Second 3DGS Challenge. In *SIGGRAPH Asia 2025, December 15–18, 2025, Hong Kong, Hong Kong*. ACM, New York, NY, USA, 3 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
SA '25, Hong Kong, Hong Kong
© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

The SIGGRAPH Asia 3DGS Challenge focuses on the critical balance between reconstruction fidelity and computational efficiency, mandating a complete pipeline execution within 60 seconds on an NVIDIA RTX 4090. This constraint requires careful tuning of the optimization schedule to ensure convergence without timeout.

1.1 3D Gaussian Splatting

3D Gaussian Splatting has emerged as a highly efficient approach for novel view synthesis and 3D scene reconstruction. In contrast to traditional neural rendering pipelines that rely on implicit scene representations, 3DGS models scenes explicitly using anisotropic 3D Gaussian primitives and renders them through efficient rasterization. This explicit structure enables rapid optimization and real-time rendering, which makes 3DGS well suited for time-constrained and latency-sensitive applications.

The development of fast 3DGS reconstruction methods is particularly relevant for edge computing and mixed reality platforms. With the increasing availability of powerful consumer headsets such as the Apple Vision Pro, the Meta Quest series, and the Samsung Galaxy XR, there is a growing demand for on-device 3D reconstruction. Achieving high-quality 3DGS reconstruction within strict time and resource budgets would support real-time spatial mapping, immersive content creation, and dynamic scene understanding directly on these devices, thereby reducing dependence on cloud infrastructure and lowering end-to-end latency for interactive mixed reality experiences.

1.2 Our Approach

We present *DeadlineDinosaur*, an exploratory framework that integrates the low-level system optimizations of LiteGS [2] with the high-level scheduling logic of DashGaussian [1] and Taming3DGS [3]. Our primary hypothesis was that by scaling resolution progressively, we could fit low-frequency geometry cheaply, allowing for a significantly higher total iteration count within the 60-second window.

2 Method

Our method is a composite system designed to maximize GPU throughput and optimization density.

2.1 The Backbone: LiteGS

We selected **LiteGS** [2] as our foundation. Standard 3DGS suffers from atomic contention during gradient reduction. LiteGS addresses this via a Warp-based Rasterizer, which assigns one warp per tile to perform intra-register reduction, significantly reducing global atomic adds. Furthermore, it employs Morton Coding to spatially sort primitives in memory (Cluster–Cull–Compact), optimizing cache locality. These optimizations provide the raw speed necessary to attempt high-iteration strategies.

2.2 Frequency-Guided Resolution Scheduling

To optimize the allocation of the computational budget, we adopt the frequency-aware scaling strategy proposed by **DashGaussian** [1], which reformulates 3DGS optimization as a progressive frequency fitting problem. This approach is grounded in the observation that downsampled images lack high-frequency residuals present in their high-resolution counterparts. Therefore, optimizing at lower resolutions is equivalent to fitting the model to low-frequency scene components.

We initialize the schedule by computing the Fast Fourier Transform (FFT) of the training images to analyze their spectral energy distribution. Let $I \in \mathbb{R}^{H \times W}$ be a training image and $F = \mathrm{DFT}(I)$ be its frequency map. We define the significance of frequency components at a specific resolution scale (h, w) as the cumulative spectral energy:

$$\mathcal{X}(F) = \frac{1}{N} \sum_{n=1}^{N} \sum_{i=1}^{h} \sum_{j=1}^{w} ||F^n(i,j)||_2 \tag{1}$$

where N is the number of training views and F^n corresponds to the n-th view. This metric $\mathcal{X}(F)$ quantifies the information content capturable at a given resolution.

To construct the schedule, we define a set of monotonically increasing downsampling factors $r_1, r_2, \ldots, r_m$. The transition iteration s_{r_i}, at which the rendering resolution increases from $(H/r_i, W/r_i)$ to $(H/r_{i-1}, W/r_{i-1})$, is determined by the ratio of spectral energy between the resolutions. Specifically, we distribute the total optimization steps S based on a logarithmic modulation of the spectral energy differences:

$$s_{r_i} = S \cdot \left(1 - \frac{\ln(\mathcal{X}(F)/\mathcal{X}(F_{r_i}))}{\ln(\mathcal{X}(F)/\mathcal{X}(F_{r_m}))}\right) \tag{2}$$

where $\mathcal{X}(F)$ represents the full-resolution spectral energy and $\mathcal{X}(F_{r_i})$ represents the energy at downsampling factor r_i. We set the initial maximum downsampling factor r_m such that $\mathcal{X}(F_{r_m}) \approx \frac{1}{a}\mathcal{X}(F)$, with $a = 4$ as the default hyperparameter. This schedule ensures that the model dedicates optimization steps proportional to the "significance" of the frequency components being introduced at each scale, progressively refining fine details only after the coarse geometry has stabilized.

2.3 Hybrid Primitive Budgeting

To manage the growth of primitives during this multi-stage process, we implement a two-phase budgeting system.

Phase 1: Momentum-Based Growth. During the variable resolution phase, we estimate the required primitive count dynamically using the momentum formulation adapted from DashGaussian:

$$P_{fin}^{(i)} = \max(P_{fin}^{(i-1)}, \gamma \cdot P_{fin}^{(i-1)} + \eta \cdot P_{add}^{(i)}) \tag{3}$$

Here, $P_{fin}^{(i)}$ represents the estimated final primitive budget at step i, and $P_{add}^{(i)}$ is the count of primitives naturally suggested for densification by the split/clone heuristics. We utilize a decay factor $\gamma = 0.98$ and an injection factor $\eta = 1.0$. This formulation allows the target budget to adaptively increase when the optimization signal (P_{add}) is strong, but decay when the geometry stabilizes.

Phase 2: Fixed-Target Refinement. Upon reaching full resolution, we transition to the **Taming3DGS** framework. We lock the budget $P_{target} = P_{fin}^{(t)}$ and switch to a score-based densification criterion. This ensures that in the final seconds of training, the system focuses on refining the attributes of the existing primitives rather than allocating new ones, preventing late-stage memory fragmentation.

3 Optimization Strategy

A key contribution of our work was the empirical tuning of the training loop to "squeeze the lemon"—maximizing the work done within the 60-second limit.

Iteration Density: By starting at 1/8 resolution, the rasterization cost in the early epochs is negligible. This allows us to perform optimization steps at a much higher frequency compared to a full-resolution baseline. We configured our schedule to execute approximately 30% more total iterations than the vanilla baseline within the same time window. Our hypothesis was that this increased "iteration density" would help the model converge faster on high-frequency details once the resolution scaled up.

4 Results and Discussion

We evaluated our method on the official challenge dataset comprising 13 diverse scenes. All local experiments and final challenge submissions were conducted on a single NVIDIA GeForce RTX 4090. Figure 1 illustrates the detailed scene-wise PSNR performance of our method across these 13 publicly available scenes, demonstrating the consistency of our reconstruction quality. Some sample renders for selected scenes can also be seen in Figure 2.

4.1 Challenge Performance

As per the final leaderboard, *DeadlineDinosaur* secured the 2nd Rank overall.

Table 1 summarizes our final standing. Our method achieved a PSNR of 28.04, staying strictly within the time limit at 57 seconds. This result validates the robustness of our hybrid scheduling strategy on the blind test set.

Table 1: Final Test Set Metrics (Official Leaderboard).

Team Name	PSNR (Avg.)	Time (s)
1. 3DV-CASIA	28.43	55
2. DeadlineDinosaur (Ours)	**28.04**	**57**
3. MT-AI	27.58	34

4.2 Comparative Analysis

To dissect the contributions of our scheduling logic, we compared our method against a vanilla LiteGS baseline configured for the same time budget on our local validation set.

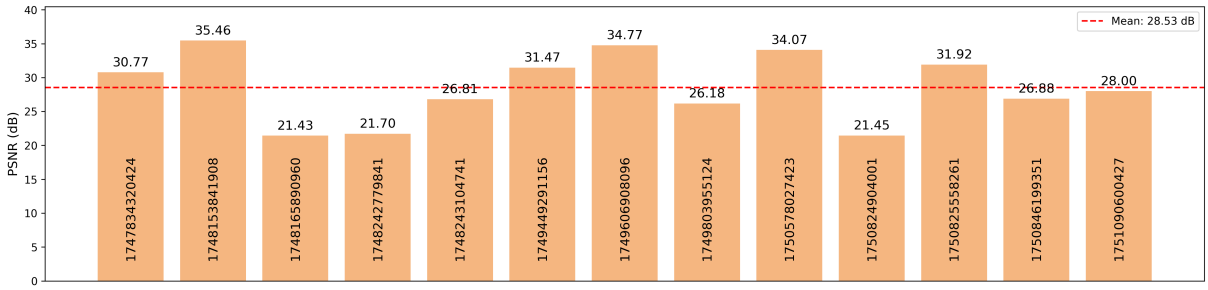

Figure 1: Scene-wise PSNR comparison across the publicly challenge dataset. The red dotted-line indicates the average PSNR.

Figure 2: Qualitative comparison of ground truth (left) and our rendered results (right) for selected scenes from the challenge dataset.

Table 2: Local validation metrics across challenge scenes.

Method	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
Vanilla LiteGS	28.54	0.882	0.158	48.23
Ours	28.53	0.874	0.168	55.19

As shown in Table 2, our method achieves performance parity with the baseline (PSNR 28.53 vs 28.54). We observed that the LiteGS rasterizer is highly optimized, such that the cost of rasterization is not the dominant bottleneck on the RTX 4090 for these scenes. Consequently, the computational savings from rendering at 1/8 resolution were less significant than anticipated. While we successfully executed more iterations, the overhead of the scheduling logic and the delayed exposure to high-frequency details (due to low-res training) balanced out the gains in raw speed.

However, this comparison highlights a critical trade-off. While vanilla LiteGS is faster on average, it relies on a reactive densification strategy that can lead to unpredictable resource usage. Our method, by enforcing a deterministic schedule derived from DashGaussian and Taming3DGS, ensures a consistent and predictable optimization trajectory. The successful deployment and high ranking on the challenge server confirm that this stability is a decisive factor in blind evaluation scenarios, where robustness often outweighs marginal gains in peak throughput. The transition to Taming3DGS in the final phase ensured that our primitive counts remained controlled, validating the momentum-based budgeting as a reliable estimator of scene complexity.

5 Conclusion

We presented *DeadlineDinosaur*, a solution for the SIGGRAPH Asia 3DGS Challenge that explores the intersection of system-level and algorithmic optimization. By layering DashGaussian's resolution scheduling and Taming3DGS's budgeting atop the LiteGS backend, we created a pipeline that maximizes iteration density. While our analysis reveals that resolution scaling offers diminishing returns when the underlying rasterizer is sufficiently fast, our framework successfully meets the challenge requirements with competitive fidelity. Achieving Rank 2 validates that carefully orchestrated scheduling is a viable path for high-performance 3D reconstruction under strict constraints, offering a reliable blueprint for future real-time optimization systems.

References

[1] Youyu Chen, Junjun Jiang, Kui Jiang, Xiao Tang, Zhihao Li, Xianming Liu, and Yinyu Nie. 2025. DashGaussian: Optimizing 3D Gaussian Splatting in 200 Seconds. In *CVPR*.

[2] Kaimin Liao, Hua Wang, Zhi Chen, Luchao Wang, and Yaohua Tang. 2025. LiteGS: A High-performance Framework to Train 3DGS in Subminutes via System and Algorithm Codesign. arXiv:2503.01199 [cs.CV] https://arxiv.org/abs/2503.01199

[3] Saswat Subhajyoti Mallick, Rahul Goel, Bernhard Kerbl, Markus Steinberger, Francisco Vicente Carrasco, and Fernando De La Torre. 2024. Taming 3DGS: High-Quality Radiance Fields with Limited Resources. In *SIGGRAPH Asia 2024 Conference Papers (SA '24)*. Association for Computing Machinery, New York, NY, USA, Article 2, 11 pages. doi:10.1145/3680528.3687694